

Programming Languages Evaluation Criteria

Readability: Programs in the language should be easy to read.

Writability: Programs in the language should be easy to write: the programmer's time is valuable.

Reliability: It should be easy to develop programs that work the way you think they will, and hard to introduce accidental bugs.

Factors affecting reliability include:

- **Do language features invite the programmer to make mistakes?**

C/C++ example:

if (x = 4) ...

- Checks for **type compatibility**, **array bounds**, and similar constraints?
- **Exception-handling**

Efficiency: How fast do programs in the language run?

How much work is required **behind the scenes**?

Postponing actions from compile time to **run time** costs speed.

Tradeoff with reliability (*Example: Array bounds checking at run time*)

Simplicity: Does the language have a lot of features, or just a few?

(How many different control constructs, data structures, operators?)

More simplicity:

Easier to learn and master the language.

May be harder to accomplish the tasks you want, with limited tools.

More complexity:

Harder to learn and master the language. In some cases, programmers may settle for learning only a subset of the language.

May be easier to accomplish the tasks you want, with lots of different tools.

Ease of learning

Suitability for a particular task